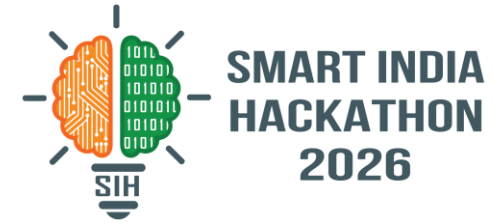


SMART INDIA HACKATHON 2026



- **Problem Statement ID – SIH26103**
- **Problem Statement Title- AI-Based
Project Monitoring and Risk Prediction**
- **Theme- AI / Digital Transformation**
- **PS Category- Software**
- **Team ID-**
- **Team Name- Visionary6**



AI-Based Project Monitoring and Risk Prediction

❖ Proposed Solution

- AI-powered platform for real-time project monitoring.
- Tracks project progress, deadlines, resources and task completion.
- Predicts possible delays, budget overruns and resource risks.
- Provides an early-warning system with actionable recommendations.
- How it addresses the problem Detects deviations from the planned schedule.
- Identifies high-risk tasks before they become critical.
- Helps managers take preventive action.
- Provides a centralized project health dashboard.
- Innovation & Uniqueness Predictive rather than only status-based monitoring.
- AI-generated Project Risk Score.
- Combines historical project data + real-time progress + resource information.

TECHNICAL APPROACH



- **Technologies**

- Frontend: React.js / HTML-CSS-JavaScript
- Backend: Python+FastAPI/Flask
- AI/ML: Scikit-learn/XGBoost
- Database: MySQL/PostgreSQL
- Visualization: Power BI / Chart.js
- Optional: IoT sensors for real-time project-site data

- **Workflow**

- Project Data → Data Preprocessing → ML Risk Prediction → Risk Score → Dashboard → Alerts → Recommended Action
AI Models Regression → delay prediction Classification → risk-level prediction
Anomaly detection → unusual project behavior NLP → analyze project reports/updates

FEASIBILITY AND VIABILITY



- Uses readily available AI/ML technologies.
- Can integrate with existing project-management systems.
- Cloud-based deployment enables scalability.
- Can begin with software-only implementation.
- Potential Challenges Incomplete or inaccurate project data.
- Limited historical project datasets.
- False-positive/false-negative risk predictions.
- Integration with existing systems.
- Mitigation Data validation and preprocessing.
- Continuous model retraining.
- Human-in-the-loop verification.
- Explainable AI for transparent predictions.

IMPACT AND BENEFITS



TeamsBenefits

- ◆ Early detection of project delays
- ◆ Better resource allocation
- ◆ Reduced project failure probability
- ◆ Improved decision-making
- ◆ Reduced operational costs
- ◆ Increased project transparency
- ◆ Real-time project health monitoring

Expected Impact Reactive Management → Predictive Management

Instead of discovering a problem after a deadline is missed, the system predicts the risk before the failure occurs.

RESEARCH AND REFERENCES



Research papers on AI-based project risk prediction
Research on machine learning for project delay prediction
Research on predictive analytics in project management
Dataset sources used for model training
Documentation for the ML frameworks/APIs used